

Assignment 9

let (X, ϵ, μ) be a measure space

$$p \in (0, \infty)$$

Let $g, f \in \mathcal{M}(\epsilon)$ and $f_n, g_n \in \mathcal{M}(\epsilon) \forall n \in \mathbb{N}$

Such that $f_n \rightarrow f$ and $g_n \rightarrow g$ in the $\mu - p'$ th mean as $n \rightarrow \infty$

Then show that $af_n + bg_n \rightarrow af + gb$ in the $\mu - p'$ th mean as $n \rightarrow \infty \forall a, b \in \mathbb{R}$

What is the condition?

We need

$$\int_X |af_n + bg_n - af - gb|^p d\mu = 0$$

We start by noticing that by a simple triangle inequality

$$|af_n + bg_n - af - gb| \leq |af_n - af| + |bg_n - gb| = |a||f_n - f| + |b||g_n - g|$$

Since $p > 0$ this implies

$$\Rightarrow |af_n + bg_n - af - gb|^p \leq (|a||f_n - f| + |b||g_n - g|)^p$$

By 7.16 we then have

$$(|a||f_n - f| + |b||g_n - g|)^p \leq 2^p \cdot (|a|^p |f_n - f|^p + |b|^p |g_n - g|^p)$$

Now we simply use linearity of the integral

$$\begin{aligned} 0 &\leq \int_X |af_n + bg_n - af - gb|^p d\mu \leq \int_X 2^p \cdot (|a|^p |f_n - f|^p + |b|^p |g_n - g|^p) d\mu \\ &= 2^p \cdot \left(|a|^p \cdot \int_X |f_n - f|^p d\mu + |b|^p \cdot \int_X |g_n - g|^p d\mu \right) \end{aligned}$$

The last equality holds since $f, g, f_n, g_n \in \bar{\mathcal{M}}(\epsilon) \Rightarrow |g_n - g|^p, |f_n - f|^p \in \bar{\mathcal{M}}(\epsilon)$,

And $2^p, |a|^p, |b|^p \geq 0$ by prop 5.2.7

But $2^p, |a|^p, |b|^p$ are all just constants and since $\int_X |f_n - f|^p d\mu \rightarrow 0$ and $\int_X |g_n - g|^p d\mu \rightarrow 0$ for $n \rightarrow \infty$

Then we must have $2^p \cdot (|a|^p \cdot \int_X |f_n - f|^p d\mu + |b|^p \cdot \int_X |g_n - g|^p d\mu) \rightarrow 0$ for $n \rightarrow \infty$

And thus $\int_X |af_n + bg_n - af - gb|^p d\mu \rightarrow 0$ by the squeeze theorem